

Lab meeting notes: temperature file and HA next steps

What we have ready, and questions after Roro's reply

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1/F Meeting Room, Patrick Manson Building — 9:30–11:30

Where things stand

- ▶ Prof. Bishai asked whether we have a monthly temperature file, and to merge monthly max/min (and lag) to AMI and stroke admission risk, then produce Table 1.
- ▶ **Yes — the monthly temperature file exists** (HKO Headquarters, 2013–2023, $N = 132$ months): mean, $T_{\min}$, $T_{\max}$, plus extreme-day counts.
- ▶ Roro confirmed ICD-9, patient-level complete admissions, age bands including 65–69 and 70–74, and that ED-to-inpatient is not in this extract.
- ▶ HA outcomes are not on my account yet (wet-ink signatures / HPC). No admission results to show today.

We do have the temperature series

Hong Kong Observatory Headquarters

Monthly mean maximum and minimum temperature, 2013-2023 (N = 132 months)

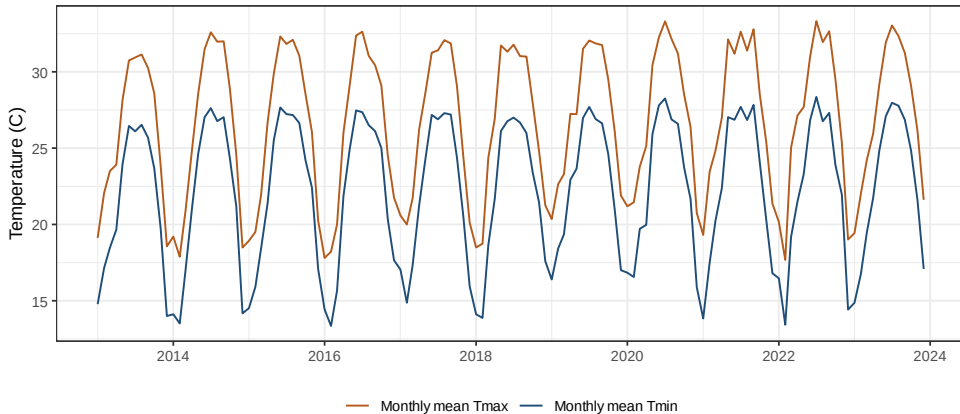


Table 1 shell — temperature filled

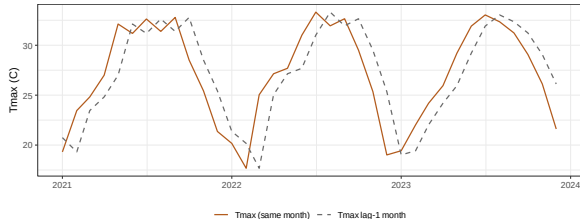
Variable (monthly, 2013–2023)	<i>N</i>	Mean	SD
Mean temperature (°C)	132	24.02	4.70
Mean Tmin (°C)	132	22.10	4.69
Mean Tmax (°C)	132	26.66	4.82
Hot nights (days/month)	132	3.40	5.40
Cold days (days/month)	132	1.10	2.55
Very hot days (days/month)	132	3.19	5.18
Age, sex	—	<i>after HA access</i>	—
Diuretics / beta blockers	—	<i>after HA access</i>	—
Metformin / SGLT2is	—	<i>after HA access</i>	—
BMI	—	<i>after HA access</i>	—

HKO Headquarters. Extreme-day annual totals also validated 33/33 against HKO *Year's Weather*. Covariate rows need the patient file.

Lag temperature — ready to construct; need the rule

Example lag construction (last 36 months)

Same-month vs previous-month mean Tmax



- ▶ Same-month Tmax and lag-1 Tmax are strongly correlated ($r \approx 0.82$).
- ▶ Easy to build lag-1 (or lag-2) of Tmax / Tmin / mean from the monthly file.
- ▶ I am unsure which lag Prof. Bishai prefers as default for the first merge.

Also already done on the public side

- ▶ Validated annual hot nights / very hot days / cold days figure.
- ▶ Exposure + aging context note (real HKO + real C&SD denominators).
- ▶ Pipeline scripts for merge once an approved HA panel is available.
- ▶ These are **not** admission associations — only preparation for the merge Prof. Bishai described.

Reading Roro's note — technical questions

1. What does **“DAE records included”** mean in this ICD-9 extract?
2. **“Complete admission outcomes at patient level”** — is that one row per admission episode, with drugs/BMI attached?
3. For **monthly risk of AMI/stroke admission**: what is the denominator / at-risk population? (All HA patients? A medication cohort? Population rates?)
4. Are **diuretics, beta blockers, metformin, SGLT2is, and BMI** already in the approved extract, or do they need a further pull?
5. How should drug exposure be timed (e.g. dispensing in the prior 90 days vs anytime before admission)?
6. Is diagnosis **principal only**, or any position? AMI and stroke codes to use under ICD-9?
7. **“No ED-to-inpatient”** — are elective inpatient admissions included?
8. On **IRB**: Roro asked whether the protocol should be updated; I will follow whatever the group decides.

What I will do next (once on the server)

1. Complete wet-ink / HPC access steps as arranged with Roro.
2. Merge monthly Tmax / Tmin (+ agreed lag) to AMI and stroke admission risk.
3. Draft Table 1 with temperature (done) and covariates (from HA).
4. Bring a first descriptive pass back to the group — not claiming causal results.

Grateful for clarification on the questions on the previous slide, especially denominator, drug/BMI fields, DAE, and lag choice.